Beliefs Beat Trees: Information Set Monte Carlo Tree Search for Exploding Kittens

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Abstract

Exploding Kittens is a small, commercially popular card game with hidden hands, a stochastic deck, a cancellation mechanic that makes turns interactive, and an elimination structure that rewards survival rather than accumulation. We present an open, dependency-free implementation of the full 56-card base game with a strict observation filter, and three agents for it: a random baseline, a rule-based heuristic, and an Information Set Monte Carlo Tree Search (ISMCTS) agent. Over 8,700 games we show that the search agent beats the heuristic policy it rolls out (score share 0.575 ± 0.017), and that essentially all of its advantage comes not from the tree but from one modelling choice: a belief distribution over where defused Exploding Kittens were buried. Removing that belief costs 0.097 of score share — more than every other component we ablated combined — and drops the agent below par against the heuristic. Measured directly, the belief cuts the Brier score of the prediction "the next card explodes" by 53% on the decisions where it has anything to say, because a buried kitten is on top of the deck 94% of the time and a uniform deck model predicts 22%. We also report a negative result and explain it: against opponents it does not model correctly, the search is *worse* than the simple heuristic it is built on, and the same belief that is worth 0.097 against modelled opponents is measurably *anti*-calibrated against unmodelled ones. On this game, at these budgets, the quality of the sampled determinizations dominated every enhancement to the tree.

Introduction

Monte Carlo tree search in games of imperfect information proceeds, in the dominant family of methods, by guessing. The agent cannot search the position it is in, because it does not have one: it has an information set, a collection of positions consistent with what it has seen. Determinization samples a member of that set, searches it as though it were the truth, and repeats; Information Set MCTS (Cowling et al., 2012a) shares a single tree across the samples so that no individual guess decides anything. The literature that grew around this family is largely a literature about the tree — selection rules, expansion policies, priors, playout policies, information reuse (Browne et al., 2012; Powley et al., 2014; Świechowski et al., 2023).

This paper is about the other half. A search over guesses is worth exactly what the guesses are worth, and we present a game in which that inequality is not merely true but overwhelming: every enhancement we made to the tree is worth, in total, less than one correction to the sampler.

The game is Exploding Kittens (Inman et al., 2015), a 56-card commercial party game. It is an unusually clean testbed for this question. Simulation is cheap: a full game between rule-based agents takes two milliseconds, so a search budget can be swept over an order of magnitude and an ablation can be run twenty-one times without a cluster. The rules are small enough to implement exactly and to test exhaustively. And yet nobody at the table ever sees a searchable position: a typical mid-game observation is consistent with roughly 10^{16} complete game states, so the uncertainty is real and sampling is forced. Cheap simulation and expensive uncertainty is precisely the combination under which one can ask *where* strength in imperfect-information search comes from, and get an answer with error bars.

Our answer, for this game, is that it comes from one publicly observed but non-uniform event. When a player draws an Exploding Kitten and defuses it, they choose where in the draw pile to bury it, and the whole table is told that this happened. A sampler that then places the kitten uniformly is not merely uninformative; it is *wrong*. In our experiments, a kitten that a competent opponent has just buried is sitting on top of the deck 94% of the time, and the uniform model assigns that event probability 0.22. The search imagines game after game in which the top card is harmless and walks into the kitten with confidence. Correcting this — a per-kitten distribution over burial depth, shifted as cards come off the top and folded forward when another kitten is buried — is worth 0.097 of score share, while the largest tree enhancement we measured is worth 0.024 and two others are statistically indistinguishable from zero.

The same result has a sharp edge, which we report rather than bury. The belief is a model of how opponents bury kittens. Against opponents who bury uniformly at random it is anti-calibrated: it raises the Brier score of the same prediction by 25%. Correspondingly, our search agent, which models its opponents as heuristic players, scores 0.751 against three

random agents — *below* the 0.790 that the plain heuristic scores in the same match. Told who it is actually playing, on the same seeds, it scores 0.785. A determinizing search is only as good as its assumption about the other seats, and improving the sampler is a way of encoding an opponent model, with all the fragility that implies.

Contributions.

1. An open, tested, dependency-free implementation of the full 56-card base game with a strict observation filter and per-recipient event scoping, so accidental cheating is structurally impossible (286 unit tests).
2. A rule-based baseline that is a genuine opponent rather than a strawman: it scores 0.790 ± 0.014 against three random agents, and it is also the policy the search rolls out, so the search must beat the policy it simulates with.
3. An ISMCTS agent with a burial-belief model over hidden kitten positions, which beats that baseline at 0.575 ± 0.017 — four and a half standard errors clear of par.
4. A component ablation and two scaling sweeps over 8,700 games that isolate what each part contributes, including the parts that contribute nothing.
5. A direct measurement of the belief as a predictor, showing both why it works and, against a different opponent pool, why it fails — which mechanistically explains a documented opponent-model failure of the agent as a whole.

Related Work

Search under hidden information. Perfect Information Monte Carlo sampling — solve many fully observable guesses and vote — is old, effective, and known to be theoretically unsound: Frank and Basin (1998) named its two pathologies, strategy fusion (the search assumes it may act differently in states it cannot distinguish) and non-locality (the value of a subtree depends on how the opponent's play elsewhere reveals the world). Long et al. (2010) supplied the counterweight, identifying properties of game trees — leaf correlation, bias, disambiguation — that predict when PIMC loses little against a game-theoretic solution. Exploding Kittens sits comfortably in the benign region of that space: disambiguation is high, because the deck resolves itself as cards come off, and elimination gives strong leaf correlation. This is why determinization works acceptably here at all, and it lets us frame our contribution as improving the sampling distribution rather than escaping PIMC's pathologies. ISMCTS (Cowling et al., 2012a) is the shared-tree answer we build on directly, including the availability-count correction in the selection rule, which matters precisely when the legal move set varies between iterations — as it does here whenever a guessed hand does or does not contain the material for a combo. Determinization has been studied at length in Dou Di Zhu (Whitehouse et al., 2011) and in *Magic: The Gathering* (Cowling et al., 2012b), and Whitehouse (2014) gives the book-length treatment. The poker line (Moravčík et al., 2017; Brown and Sandholm, 2019) solves a different problem — equilibrium computation under betting — although Brown et al. (2018) arrives from the CFR side at the same idea we use below: stop the lookahead early and substitute a value.

MCTS enhancements. We adopt three standard pieces and measure all of them. Early playout termination — cut the simulation short and score the position with a static evaluation — was introduced for Amazons (Lorentz, 2008); the later study (Lorentz, 2016) shows across three games that even a weak evaluation function compares favourably to a long random playout, which is directly the trade we make. Progressive bias (Chaslot et al., 2008) supplies our decaying heuristic bonus in the tree policy. Gelly and Silver (2007) named the distinction between prior knowledge in the simulation policy and prior knowledge in the tree; we use knowledge in both places. Powley et al. (2014) give a framework (ICARUS) for describing and combining exactly this kind of enhancement, and our ablation is best read as a small instance of the empirical programme it proposes.

Agents for hidden-information card games. The closest methodological sibling is Di Palma and Lanzi (2018), who pit rule-based players encoding established human strategy against cheating MCTS and fair ISMCTS in Scopone, and find search ahead of hand-written strategy. Our protocol follows that template. It is worth stating that the opposite outcome is also on record: Malla (2025) report a rule-based aggressive agent winning 88.3% of a cross-category tournament in Dhumbal, ahead of ISMCTS at 9.0% — so it is not eccentric for a good heuristic to beat a search in this class of game, and our own negative result is less surprising in that light. Goodman (2019) is unusually relevant for a cooperative game: it combines re-determinizing ISMCTS, a learned evaluation substituted for full simulations under a 40 ms budget, and an online Bayesian opponent model — the last of which is exactly the mechanism our negative result calls for. In trick-taking games, Buro et al. (2009) is the strongest prior demonstration of this paper's thesis in another domain: learned state evaluations used to *infer* opponents' hidden cards produced measurable gains beyond search alone, with Rebstock et al. (2019) extending the inference with supervised opponent models. In digital collectible card games (Santos et al., 2017; Kowalski and Miernik, 2023) the difficulty is concentrated in branching factor and board evaluation rather than in belief, and the strongest competition entries converge independently on our design: shallow search with a heuristic cut-off. For multi-player payoffs specifically, Sturtevant (2008) shows UCT computing a mixed-strategy

equilibrium in n-player games and supports treating each seat as maximising its own scalar payoff, which is what our metric does; Nijssen and Winands (2011) study the corresponding rollout policies.

Belief and opponent modelling. Cowling et al. (2015) show inference and bluffing emerging from information already present in an ISMCTS tree; our belief is a deliberately narrow, hand-built instance of what that work does in general, and the honest comparison point. POMCP (Silver and Veness, 2010) is the principled version of what our sampler approximates — a particle belief maintained alongside the search, requiring only a black-box simulator, which our playouts are. Morenville and Piette (2025) argue for exactly the architecture we instantiate: state estimation as a separable module feeding a generic search rather than inference logic tangled into the search itself. Our result — the separable belief is worth more than any tree enhancement we tried — is empirical support for that premise. On the failure side, Ganzfried and Sun (2018) formalise the exploitation/robustness tradeoff that our mismatch experiment instantiates, and Bitan and Kraus (2017) build opponent prediction into ISMCTS and evaluate against 120 human subjects.

Exploding Kittens specifically. The thread is thin but not empty. The TAG framework (Gaina et al., 2020) includes an Exploding Kittens implementation with an action stack for Nope responses, and reports measured game statistics — branching factor 4.99, 84.5% hidden information — that we use below as an independent characterisation. PyTAG (Balla et al., 2023) trains PPO against the TAG baselines and reports win rates dropping from 0.74 ± 0.04 at two players to 0.35 ± 0.05 at four against Random, an independent observation of the table-size degradation we also measure; the follow-up (Balla et al., 2024) reports self-play PPO reaching terminal reward 0.51 ± 0.23 against Random but only 0.31 ± 0.20 against an untuned MCTS baseline in the two-player game, with wide intervals from five evaluation episodes. Patil et al. (2018) is the earliest MCTS treatment we could locate, as a course report: it formulates the two-player game as an MDP with no information-set or belief structure, and explicitly removes Nope, See the Future and the three- and five-card combos — that is, it removes the one genuinely interactive part of the game. Weiss (2016) builds an Exploding Kittens AI in a senior thesis; we cite it as evidence of prior interest without making technical claims about the method. Finally, the recent standardised card-game testbed Valet (Goadrich et al., 2026) explicitly scopes itself to traditional games played with standard decks, which excludes this class of game by construction. We are not aware of prior work that applies an information-set search with an explicit belief over hidden state to this game, that ablates the components of such an agent, or that models a publicly observed but non-uniform *placement* event of the kind a kitten burial is.

Card	#	Card	#
Exploding Kitten	$p-1$	See the Future	5
Defuse	6	Nope	5
Attack	4	Cat cards (5 kinds)	4 each
Skip	4	Favor	4
Shuffle	4		

Table 1: The 56-card base game, for p players. Kittens and spare Defuses are held back during the deal; every player receives one Defuse and seven other cards, and the remainder is shuffled together to form the draw pile.

The Game

Exploding Kittens is played with the deck of Table 1. Kittens and Defuses are held back; each player is dealt one Defuse and seven other cards; the kittens and the spare Defuses are then shuffled into the remainder, so that the draw pile always contains exactly one fewer kitten than there are players. A turn consists of playing any number of cards, in any order, and then drawing one. Drawing an Exploding Kitten eliminates the player unless they hold a Defuse; if they do, the Defuse is discarded and the player *chooses where in the draw pile to bury the kitten*, without showing anyone. The last player alive wins.

The cards are few and mostly obvious. Skip ends the turn without drawing. Attack ends the turn and hands the next player two turns. See the Future shows the top three cards to the player who played it. Shuffle randomises the draw pile. Favor takes a card of the target's choosing. The five kinds of cat card do nothing on their own and exist to be spent in combos. Nope cancels the card most recently played by anybody, may be played out of turn, and may itself be Noped.

Rule choices. Two rules differ legitimately between implementations, and results are not comparable across them, so we state ours. *Attack stacking:* an Attack ends all of the attacker's owed turns and passes on the turns still owed plus two, so attacks compound — attacking back on the first of two owed turns leaves the following player with three. *Combos:* two matching cards steal a random card, three matching cards demand a named card from a chosen player (and get nothing if the guess is wrong), and five distinct cards recover a named card from the discard pile. Any cards may be used, not only cat cards, and Defuses are excluded, since spending one is never anything but a blunder. Both choices are switchable in the implementation.

What makes it hard. Three properties matter for search. First, hands are hidden but the deck's kitten *density* is public — everybody can count how many kittens went in and how many have come out, so the game supplies a strong, correct, universally available prior that a good agent must beat rather than merely use. Second, the Nope window makes every card play interactive: a card can be cancelled by a player who is not on turn, so no action is atomic and the turn order is

not a reliable guide to who acts next. TAG's implementation models this with an explicit action stack (Gaina et al., 2020). Third, the objective is survival relative to others, not accumulation: there is no score to maximise, only an order to finish in, which makes an n-player payoff vector the natural formulation (Sturtevant, 2008).

State and information space. The state space is small; the information space is not. We measured the size of the information set directly, by counting, at each decision of 40 games between rule-based agents, the number of complete states consistent with the acting player's observation — that is, the number of ways to deal the unlocated multiset of cards into opponents' hands of known size and an ordered draw pile, computed exactly by dynamic programming over card types. The median across decisions is $10^{16.7}$; at the first decision of a four-player game it is $10^{30.3}$; it falls to $10^{3.9}$ only after turn 25, when the deck is nearly exhausted. Enumeration is not available at any point in the game that matters, which is the argument for sampling.

The move space, by contrast, is small once redundancy is removed. Over 2,532 decisions we measured a mean of 13.5 legal moves (median 4, maximum $1{,}518$: a rich hand offers hundreds of ways to buy the same effect, since a five-card combo may be assembled many ways and a three-of-a-kind may name any of twelve cards). Pruning moves that differ only in the price paid for an identical effect — keeping every distinct target and every distinct demand — reduces this to a mean of 5.4 (maximum 32), in close agreement with the branching factor of 4.99 that Gaina et al. (2020) report for an independent implementation.

Approach

Testbed and Observation Model

The implementation is separated into three layers that do not reach into each other: the game does not know who is playing, the agents do not know how a match is run, and the experiment harness knows no rules. Only the engine mutates game state.

The property that makes our comparisons meaningful is the observation filter. An agent is never handed a game state. Every decision is made from an `Observation` carrying the player's own hand, the public facts (hand sizes, discard pile, deck size, whose turn it is and how many turns they owe), and any private knowledge that player has earned with a See the Future. The structure has no field for another player's hand or for the order of the deck, so cheating by accident is not possible. Legal moves are generated *from the observation* rather than from the state, so an agent is offered exactly the list it could have derived itself. One consequence is worth spelling out: demanding a card an opponent does not hold is a legal move, because the asker cannot know. Events carry a `private_to` seat and are delivered only to the players entitled to them, so a Favor is public as "player 1 gives a card to player 2" and private, to those two players, as which card it was. Two of the 286 unit tests exist for this alone: one replays whole games and checks that the deck composition still adds up, and one checks that no agent was ever handed an event it should not have seen.

Relative to TAG (Gaina et al., 2020) and Valet (Goadrich et al., 2026), both multi-game frameworks, ours is a single-game implementation with no dependencies beyond the standard library. That is a narrower artefact, and deliberately so: a full comparison of 8,700 games, including a search agent in 7,700 of them, completes in nine minutes on a desktop, which is what made the ablation study affordable (Bard et al., 2020).

Baselines

RANDOM picks uniformly among the pruned legal moves, Nopes on a coin flip, and buries a defused kitten at a uniformly random depth.

HEURISTIC is an ordered priority list, not a search. In order: buy a Defuse if the board offers one (a five-card combo out of the discard, or a three-of-a-kind demand aimed at the largest hand); cash a pair of cat cards, which do nothing in hand; escape a kitten it can see on top, preferring Attack or Skip over Shuffle so that the next player inherits the problem; take a draw it knows to be safe; buy a See the Future once the deck is thin enough that the answer could change its mind; and duck out when the odds are bad, with the risk threshold conditioned on whether it holds a Defuse (0.35 with, 0.12 without), because with a Defuse the disaster becomes a weapon. Its burial rule is to place a defused kitten exactly as deep as its own remaining owed draws — the shallowest position it survives, which usually means directly on top for the next player. An ε parameter makes it depart from the rules with small probability; this is used in rollouts, where a deterministic policy would make every playout from the same guess identical.

This is a real opponent, not a strawman: it scores 0.790 against three random agents. It is also the rollout policy and the opponent model of the search agent, so the search must beat the policy it simulates with — a sharper test than beating random play, and the same design Di Palma and Lanzi (2018) use.

Information Set MCTS

MCTS is HEURISTIC with a better answer to the one question worth thinking hard about. Only the turn decision is searched; Nope responses, which card to give for a Favor, and where to bury a kitten are inherited from the heuristic. The four phases are instantiated as follows.

Selection. Descend by UCB1 (Auer et al., 2002; Kocsis and Szepesvári, 2006) while every move on offer has a child. Because each iteration deals different cards, the moves on offer at a node change between iterations — a combo exists only when the guessed hand contains the material — so the

Algorithm 1 One decision of the search agent

$C \leftarrow$ PRUNE(LEGALMOVES(o))
if $|C| = 1$ **then return** the single move
end if
$T \leftarrow$ empty tree with root r
for $i = 1$ **to** *iterations* **do**
  $\mathbf{d} \leftarrow$ BELIEFPOSITIONS(o) ▷ depth per defused kitten
  $s \leftarrow$ DETERMINIZE(o, *tracker*, $\mathbf{d}$) ▷ kittens never in hands
  $n \leftarrow r$; *inTree* $\leftarrow$ **true**
  play s forward with the real engine, where
    our turn decisions, while *inTree* and above the
    depth limit, come from DESCEND($n, \cdot$);
    all other decisions come from the heuristic policy
  stop after *maxPlayoutTurns* further turns
  $v \leftarrow$ POSITIONVALUE(s, *self*)
  back up v along the visited path
end for
return the root child with the most visits

exploration term is divided by the number of iterations in which a move was *available*, not by the parent visit count (Cowling et al., 2012a). The count is kept on the parent, one entry per move it has ever offered, so a move offered before it was first expanded is credited for those iterations too.

Expansion. The first time an offered move has no child, add one and stop descending.

Simulation. Play the guessed state on with the heuristic policy, with ε noise, on every seat, and stop after 8 further turns. Playouts run through the real engine rather than a re-implementation, so a simulation cannot silently disagree with the game it is advising on.

Backpropagation. Score the reached position in $[0, 1]$ and add it along the path.

The tree covers one of the searching player's own decisions by default; below that the heuristic takes over. The limit is deliberate, and we return to it in the results: a node the search has visited three times picks our move from a three-playout estimate, which is barely better than picking at random, and every root value measured through such a node is a value measured under bad play by ourselves.

Determinization and the Kitten Belief

This is the core of the paper. A determinization is a complete game state consistent with the observation: the observing player's hand, the discard pile, the hand sizes, whose turn it is and how many turns they owe are facts and are copied across; the unlocated cards — the deck plus every opponent's hand — are dealt out at random. Cards the player has seen on top of the deck with a See the Future stay on top, in order. Three corrections turn this from a sampler that loses games into one that wins them.

Kittens are never in a hand. A kitten in the draw pile is public information: it is counted, and it cannot be in anybody's hand, because drawing one either kills you or is announced. Dealing the unlocated pool out naively ignores this and hides kittens in opponents' hands. The size of the error is easy to measure: averaged over decisions, the draw pile is 58% of the unlocated pool, so a naive sampler places only 58% of the kittens in the deck and the imagined deck is about 40% safer than the real one. Constraining kittens to the draw pile costs nothing and removes the bias exactly.

The burial belief. The kittens shuffled in at the start of a game are uniformly placed, and counting them is all anybody can do. A kitten that has been *defused* is a different matter: a player drew it, chose a depth, and the whole table was told that it happened. Sampling its position uniformly does not merely throw information away — it is wrong in a specific and costly direction.

KITTENBELIEF keeps one distribution over depth per defused kitten. When a kitten is buried into a draw pile of m cards, the $m + 1$ open positions receive

$$P(d) \;=\; w\,\frac{\gamma^d}{\sum_{j=0}^{m}\gamma^j} \;+\; (1-w)\,\frac{1}{m+1}, \quad d = 0, \ldots, m,$$

a mixture of a geometric preference for shallow positions with a uniform component. The two constants are hand-set: $w = 0.7$ (weight on the assumption that a rational burier puts the kitten where the *next* player will meet it) and $\gamma = 0.5$ (how fast that assumption decays with depth). When the searching player buries a kitten itself, the distribution is a point mass at the depth it chose.

The belief is then maintained as the game proceeds. When a card comes off the top without a kitten being reported, every distribution is shifted up one and renormalised — which is a Bayesian update conditioning on the kitten *not* having been the card drawn. When a kitten is drawn, the tracked distribution that best explains it is retired. When a new kitten is buried, it is an insertion, so every kitten already tracked moves down one position with probability equal to the mass of the insertion distribution at or above it; we apply this convolution exactly. A Shuffle wipes the belief entirely, since nobody knows anything about the deck any more. To sample, the tracked kittens are placed first, sharpest belief first, and any remaining kitten is placed uniformly among the free positions — which is the honest answer for a kitten nobody has touched.

Two things are worth being explicit about. First, this is a belief over a publicly announced but privately parameterised *placement*, which is an unusual shape for a card-game belief: the event is common knowledge and only its parameter is hidden. Second, it is a separable module — the search calls it for a list of depths and is otherwise unaware of it — which is the architecture Morenville and Piette (2025) argue for on general grounds.

Hand tracking. Some hidden cards are not really hidden. When this player hands a card over for a Favor, or a three-of-a-kind takes a named card off it, the player is told what moved. HANDTRACKER follows those events and deals such

cards back to the opponent holding them, forgetting a tracked card as soon as its holder could have got rid of it, so the picture never claims a card an opponent no longer has. We report in advance that this did not measurably help.

Truncated Playouts and Position Evaluation

A playout run to the end of the game answers the right question and answers it terribly. Thirty turns of stochastic deck and three opponents sit between the move under test and the outcome, while the candidate moves differ in value by a couple of percent; averaging enough such playouts to see two percent takes hundreds of samples *per move*, which is not a budget this game affords.

Playouts are therefore cut off after eight turns and the reached position scored by an evaluation function (Lorentz, 2008, 2016). The evaluation returns the same quantity a finished game would — this player's share of the placement points — so truncated and completed playouts are directly comparable and the search never needs to know which it got. It is exact whenever the game is decided (an eliminated player has a placement; a lone survivor has won). Otherwise it computes, for each surviving player p, a log-odds strength

$$\sigma(p) = 1.2\, n_{\mathrm{def}} + 0.35\, n_{\mathrm{esc}} + 0.2\, n_{\mathrm{nope}} + 0.03\, |h| - 2.0\, \delta,$$

counting Defuses (a life), Skips and Attacks (a dodged draw), Nopes, hand size (combo fuel), and exposure δ: kitten density times the draws the player cannot avoid, discounted to a quarter if they hold a Defuse, and using the actual top card rather than the density for the imminent draw, which inside a determinization is a concrete fact. The survivors' softmax shares of σ interpolate each player between the best and worst finishes still open to them. The weights are hand-set, not learned. Lorentz (2016) report that even a weak evaluation compares favourably to a long random playout, which is the premise we are relying on.

Prior Knowledge in the Tree Policy

The move the heuristic would play is expanded first, and thereafter carries a bonus of $\beta/(v+1)$ in the selection rule, where v is the child's visit count: progressive bias (Chaslot et al., 2008), with $\beta = 1$. The motivation is the same variance argument as above — with 120 playouts split over five candidate moves, the search cannot separate two moves whose true values differ by a percentage point, and without a prior its choice is decided by that noise. With one, it starts from the heuristic's move and needs evidence to leave it. Together with the heuristic rollout policy, this is prior knowledge in two of the three places Gelly and Silver (2007) identify.

Experimental Setup

Metric. Everything is reported as **score share**: mean placement points over the maximum, where a player finishing k-th of p scores $p - k$ points, so 1.0 is winning every game, 0.0 is going out first every time, and equal agents score exactly 0.5 at any table size. We prefer it to win rate for three reasons: it is comparable across table sizes, which matters because we sweep them; it gives credit for surviving to second place, which is most of what skill buys in an elimination game; and it has a fixed null value, which makes any real difference legible. It is the natural scalarisation of the per-player payoff vector that n-player search assumes (Sturtevant, 2008). We are not aware of a prior name for this quantity in the literature, so we state it as our definition. Standard errors are reported throughout and we treat two of them as the threshold for a claim.

Parameter	Value	Meaning
iterations	120	playouts per decision
exploration	0.4	UCB1 constant, rewards in $[0, 1]$
bias β	1.0	weight of the heuristic prior
rolloutEpsilon	0.15	rollout noise
treeDepth	1	own decisions inside the tree
maxPlayoutTurns	8	turns before cut-off
useEvaluation	on	score cut-off playouts
useBelief	on	burial belief
useTracker	on	hand tracking
opponentModel	heuristic	who the search assumes it plays
w, γ	0.7, 0.5	burial shallowness and decay

Table 2: Default search configuration. At these settings a decision costs about 45 ms on one core, and a four-player game about 0.5 s.

Protocol. Seating rotates by one seat per game, so that turn-order advantage — which is real in this game — cannot decide a comparison; a per-seat breakdown serves as the check that the rotation covers everything. A single seed drives both the shuffles and every agent's decisions, so any batch replays exactly, and matches that differ only in an agent option (such as the opponent-model comparison) are run on identical seeds and are therefore directly paired. The full comparison is 21 matches and 8,700 games, run in parallel across cores, and writes a machine-readable results file containing every per-agent statistic. It takes 516 seconds. Correctness scaffolding is 286 unit tests over the deck, legal move generation, the observation filter, every card effect, the belief, the evaluation, aggregation and reproducibility.

Experiments. Four families. (i) *Head-to-head strength*: the search against three random agents and against three heuristic agents, both baselines against each other, a mixed table with one of each, and a random-only control that must return 0.500 if the harness is sound; 500 games each. (ii) *Component ablation*: each part switched off in turn against three heuristic opponents, 400 games each, read against a full-agent reference run on the same seeds. (iii) *Budget sweep*: 15 to 240 playouts per decision, 400 games each. (iv) *Table size*: two to five players, 300 games each.

Separately, and outside the match harness, we measured the belief directly as a predictor. For each turn of 400 games we recorded the probability that the acting player's next card is an Exploding Kitten under two models — the public uni-

Match (500 games each)	Score share
four RANDOM agents (control)	0.500 ± 0.008
HEURISTIC vs. three RANDOM	0.790 ± 0.014
MCTS vs. three RANDOM	0.751 ± 0.014
MCTS vs. three RANDOM, matched model	0.785 ± 0.015
MCTS vs. three HEURISTIC	$\mathbf{0.575 \pm 0.017}$
Mixed table: one of each plus a second RANDOM	
MCTS	0.751 ± 0.015
HEURISTIC	0.591 ± 0.015
RANDOM	0.329 ± 0.010

Table 3: Head-to-head results. Par is 0.500.

form density k/D, and the belief, estimated by sampling — and compared both with the truth, taken from the real state after the fact.

Results

Head-to-Head Strength

Table 3 gives the headline. The control returns 0.500 ± 0.008, confirming that the harness, the seat rotation and the metric are sound. The heuristic beats random play by a wide margin. The line that matters is the last: the search beats three copies of the policy it rolls out at 0.575 ± 0.017, four and a half standard errors clear of par. Searching is worth doing, and the gain is not an artefact of a weak rollout policy, because the rollout policy is the opponent. The mixed table settles the ordering within the same games, with the search at 0.751, the heuristic at 0.591 and random play at 0.329.

The Negative Result

Against three random opponents the search scores 0.751, *below* the 0.790 that the plain heuristic scores in the same match. Told to model its opponents as random players instead of as heuristics — the only change, on identical seeds — it scores 0.785 ± 0.015. The wrong opponent model costs 0.034, and the residual gap to the heuristic is then within noise.

This is the expected behaviour of a determinizing search (Ganzfried and Sun, 2018; Long et al., 2010), and we report it plainly. It is not the whole story, though, and the belief measurement below explains *why* the mismatch bites in this particular game: the burial belief is itself an opponent model, and against opponents who bury uniformly at random it is not merely uninformative but actively misleading.

Ablation

Table 4 is the paper's central experiment. Removing the kitten belief costs 0.097 of score share and drops the agent to 0.484 — below par, i.e. worse than the heuristic opponents it is playing. That single component is worth more than the agent's entire margin over the heuristic, and more than everything else in the table put together. Truncated-and-evaluated playouts are worth 0.024, a real but much smaller

Configuration	Share	Δ
Full agent (reference)	0.581 ± 0.019	—
No kitten belief	0.484 ± 0.020	$\mathbf{-0.097}$
Full playouts, no evaluation	0.557 ± 0.019	-0.024
No heuristic prior ($\beta = 0$)	0.573 ± 0.019	-0.008
Random opponent model in playouts	0.584 ± 0.018	$+0.003$
Tree depth 4 instead of 1	0.603 ± 0.019	$+0.022$
No hand tracker	0.611 ± 0.018	$+0.030$

Table 4: Ablation against three HEURISTIC opponents, 400 games each on matched seeds. Δ is the change in score share from switching the component off. Only the first two rows are resolved at this sample size.

	Opponents bury as	
Quantity	HEURISTIC	RANDOM
Turns with a tracked burial	924 / 2984	1538 / 3589
Kitten actually on top	0.943	0.324
Predicted, uniform deck	0.220	0.334
Predicted, belief	0.450	0.486
Brier score, uniform deck	0.599	**0.160**
Brier score, belief	**0.284**	0.201
change	-53%	$+25\%$

Table 5: The belief measured as a predictor of "the acting player's next card is an Exploding Kitten", on turns where at least one burial is being tracked, over 400 games per column. Lower Brier is better. On turns with nothing tracked the belief reduces exactly to the uniform model, and the two scores agree to within 0.3%.

effect, and the only tree-side enhancement that clears the noise floor.

The remaining four rows are, at 400 games, indistinguishable from zero, and we state that rather than rounding them in our favour. The heuristic prior is worth a nominal 0.008; the standard error on a difference of two matches at this sample size is roughly 0.027. More awkwardly, two components are nominally *negative*: switching off the hand tracker gains 0.030 and deepening the tree from one own-decision to four gains 0.022. We do not claim these as real effects — both are around one standard error — but we do not hide them either, and we note that neither has an obvious mechanism that would make it surprising if it were real. A deeper tree at a fixed 120-playout budget spreads the same evidence over more decisions; hand tracking sharpens the guessed hands but not the guessed deck, and in this game the deck is what kills you. Resolving effects of this size would need on the order of 10^4 games per condition by brute force, or a variance-reduced sequential protocol (Li et al., 2026).

Why the Belief Works, and When It Does Not

Table 5 explains the ablation. Against heuristic opponents, on the 31% of turns where the belief is tracking at least one burial, the top card of the deck really is an Exploding Kitten 94% of the time — because a competent player who defuses a kitten puts it exactly where the next player will draw it.

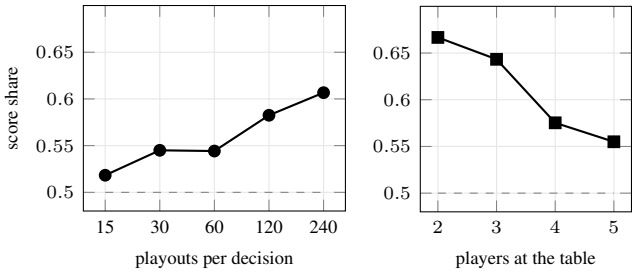


Figure 1: Left: score share against three HEURISTIC opponents as the playout budget varies (400 games per point). Right: score share against HEURISTIC opponents as the table grows (300 games per point, four players from Table 3). Dashed line is par.

The public uniform model, which is the best an agent can do from card counting alone, assigns that event probability 0.22. It is not slightly wrong; it is confidently wrong in the direction that gets a searcher killed. The belief moves the prediction to 0.45 and cuts the Brier score by 53%. Over all turns, including those where nothing is tracked, the reduction is 34%.

It also shows the limit. The belief is still badly underconfident: 0.45 against a base rate of 0.94. Its constants were set by hand and never fitted, and this measurement says that fitting them — or replacing the fixed shape with a burial policy inferred from observed outcomes — has substantial room left, which is the first item of future work below.

And it shows the failure mode. Random opponents bury uniformly, so against them the uniform model is correct by construction (0.324 actual against 0.334 predicted) and the belief, insisting on shallowness, is anti-calibrated: its Brier score is 25% *worse*. This is the same 0.034 opponent-model mismatch of Table 3 seen at the level of the prediction that causes it. The general lesson is uncomfortable but clear: improving a sampler by modelling how opponents behave imports the opponents into the sampler, and a better sampler against one population is a worse one against another.

Scaling

Figure 1 (left) sweeps the budget. Score share rises from 0.518 at 15 playouts per decision to 0.607 at 240, with no sign of saturation at the top of the range we could afford; the 30- and 60-playout points are level with each other, but the trend across the range is monotone within noise. The reported strength of the agent is therefore a budget choice rather than a ceiling. It is worth noting what the low end means: at 15 playouts the agent is still above par, which says that most of the value is not in resolving move choices at all but in the priors and the sampler that a single playout already carries.

Figure 1 (right) sweeps the table size. The advantage is largest head-to-head (0.667) and shrinks steadily to 0.555 at five players. This is the expected dilution — more opponents means more hidden cards per guess, more of the outcome outside any one player's control, and a shorter game per seat (mean turns survived falls from 13.1 at two players to 5.4 at five). The direction agrees with Balla et al. (2023), who observe a much sharper drop for PPO on the same game in a different implementation, from a 0.74 win rate at two players to 0.35 at four. The metrics are not comparable and the algorithms are not the same; the agreement is in the sign.

	Boom%	Turns	Cards	Defuses
vs. three HEURISTIC				
MCTS	67.2	7.69	9.25	1.78
HEURISTIC	77.6	7.20	8.20	1.81
vs. three RANDOM				
MCTS	45.2	8.72	10.31	1.51
RANDOM	84.9	7.43	9.37	1.25

Table 6: Behavioural counters, per seat per game. "Boom%" is the share of games the agent exploded in; "Turns" is turns taken before elimination.

How the Search Wins

Table 6 makes the mechanism concrete. The search does not win by out-drawing anybody; it wins by not exploding. Against heuristic opponents it explodes in 67.2% of games against their 77.6%, survives half a turn longer, and plays a card more per game — it converts more of its hand into tempo and steals while spending no more Defuses. Against random opponents the gap is wider (45.2% against 84.9%), which is the same behaviour against a weaker field. The pattern is consistent with the belief being the active ingredient: the counter that moves is the one that a correct estimate of "does the next card explode" should move.

Threats to Validity

We report results from a single implementation of a game whose rules admit legitimate variation, and our choices on attack stacking and combo legality are stated in the rules section precisely because results are not comparable across them. The evaluation weights are hand-set rather than learned, though Lorentz (2016) suggest a weak evaluation suffices for this purpose. The opponent pool is three agents; there is no human play, and no calibration against how a person would rate the agents' play. We do not run a cheating, perfect-information MCTS as an upper bound, which Di Palma and Lanzi (2018) do and which is the most likely reviewer request; the reason is scope rather than principle, and it would bound how much of the remaining gap is attributable to the sampler at all. Ablations at 400 games cannot resolve effects below roughly 0.03, which leaves four of six ablation rows unresolved; a variance-reduced sequential protocol (Li et al., 2026) is the principled fix and would be cheaper than brute force. Finally, and most importantly for interpretation, the belief calibration in Table 5 is measured against specific burial policies. Its headline number is conditioned on opponents

that bury shallowly; that is the same population as the headline match, so the comparison is internally consistent, but it means the 0.097 ablation result should be read as "the belief is worth 0.097 against opponents whose burial behaviour it models correctly" rather than as a property of the game alone. The random-burier column is the evidence for that reading.

Conclusion and Future Work

On this game, at these budgets, the quality of the sampled determinizations dominated every enhancement we made to the tree. One correction to the sampler — a belief over where a defused Exploding Kitten was buried — was worth 0.097 of score share, more than the agent's whole margin over the policy it rolls out, while progressive bias, a deeper tree and hand tracking were individually indistinguishable from zero at 400 games and truncated evaluation was worth 0.024. Measured directly as a predictor, the belief cuts the Brier score of the only prediction that matters in this game by 53% on the turns where it applies.

The transferable lesson is about the direction of the error, not its size. Before tuning a search over sampled worlds, audit whether the sampler is *wrong* rather than merely uninformed. An uninformative prior wastes information; a wrong prior actively misleads, and a search is an amplifier for that. In Exploding Kittens the wrong prior was the obvious one — uniform over deck positions — and it was wrong because it ignored a decision that a player made, in public, for a reason. Games are full of such events.

Future work, roughly in order of expected value. *Infer the burial depth from behaviour* rather than assuming it: the belief's constants are hand-set and, per Table 5, under-confident by a factor of two against the opponents it does model, so fitting the shape to observed outcomes should pay immediately (Buro et al., 2009). *Online opponent modelling* to fix the documented mismatch: classify opponents from their play and switch both the rollout policy and the burial prior, or maintain a mixture over opponent types (Goodman, 2019; Bitan and Kraus, 2017; Rebstock et al., 2019). Our two results — the 0.034 mismatch cost and the anti-calibrated belief against random buriers — are a measurement of exactly what such a model would buy. *A learned evaluation* in place of the hand-weighted position value, trained on self-play outcomes (Gelly and Silver, 2007). *A belief over the whole deck*, not just kittens, via a particle filter weighting determinizations by how consistent observed actions are with them; POMCP (Silver and Veness, 2010) is the algorithm this describes. *Bluff and Nope timing as search decisions*: these are currently delegated to the heuristic, and the Nope window is the one genuinely interactive, deception-capable part of the game (Cowling et al., 2015). *Deeper trees at larger budgets*: the depth limit is nominally unhelpful at 120 playouts and the budget sweep has not saturated, so the two should be swept jointly. *Rule variants* the implementation already supports — nastier decks, more kittens, no attack stacking — as a robustness study. And *human evaluation*, to check that score share against these baselines corresponds to anything a person would recognise as strong play.

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